



Ethylene bis-carbonates as telltales of SEI and electrolyte health, role of carbonate type and new additives



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ABSTRACT

The ethylene bis-carbonate compounds formation is responsible for the earliest change in electrolyte composition which can be one of the reasons for battery performance decay. In this study, liquid GC/MS technique is used to detect their formation in electrolytes based on solvent mixtures of EC and different linear carbonates (DMC, DEC and EMC), after the first cycle in full cells composed of synthetic graphite powder/commercial positive films. These compounds stem from linear carbonate electrochemical reduction leading to alkoxide compounds and can be quantified using a selective bicyclic boron ester Lewis acid as an electrolyte additive. Moreover, a quantitative study on ethylene bis-carbonate compounds for which the generation profile is different depending on the linear carbonate type, shows that either in batteries or in a simple chemical mixture of electrolyte and lithium alkoxide, their formation stops when it reaches a threshold concentration due to the thermodynamic equilibrium. The overall information is useful for investigating the passivation ability and the dissolution of the Solid Electrolyte Interphase (SEI) that is formed on the negative electrode material. Finally, the passivation property of the SEI freshly formed with four additives - Vinylene Carbonate (VC), Vinyl Ethylene Carbonate (VEC), Fluoro Ethylene Carbonate (FEC) and 1,3-Propane Sultone (1,3-PS)- is studied.

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1. Introduction

The Li-ion batteries (LIBs) have become, since the last decade of the 20th century, the most popular type of rechargeable battery. Initially common for consumer and portable electronics, the LIBs are also more and more used in larger scale applications such as aerospace and automotive industry.

As a response to the expanding range of application for LIBs, research has been constantly carried out to improve the cycle life performance, especially for large sized battery units. Studies of electrolyte degradation remain the main key for this research, since it is inevitable as the stability window of the electrolyte is exceeded, principally at the negative side. Such degradation can be more or less limited thanks to the passivation layer formed at the surface of negative active material during the initial charging of the battery. The properties of this passivation layer called SEI (Solid Electrolyte

Interphase) [1] depend on several factors, such as the chemical nature of solvents and electrodes, the charging protocol and the temperature.

A lot of studies have already been conducted over the chemical nature of SEI layer since the late 1980's starting from Li metal batteries [2]. These studies, using mainly FTIR (Fourier-Transformed Infra-Red) and XPS (X-ray Photoelectron Spectroscopy) [3], resulted from the logical approach that consists in linking the SEI chemical nature with its passivation properties. However, this approach is questionable considering the huge diversity of SEI components despite its extreme thinness (usually dozen of nanometers) and the probable existence of composition gradients across it that make the analysis of the compounds difficult, and several techniques need to be combined in order to obtain accurate information [4,5]. The passivation properties of this layer remain unclear even though its constituents are finally known.

Another approach is to analyse soluble compounds that are generated concomitantly with SEI layer [6], to have an overview on the whole electrolyte degradation process, and thus better understand the battery performance decay over time. Indeed, some soluble or solid degradation compounds may eventually

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EC/DMC and EC/DEC-based electrolyte at a known concentration. Integrated intensities of each peak were compared to that of EC. The intensity ratio thus obtained was converted into the relative molar quantity of each bis-carbonate versus EC. Knowing the EC content in each electrolyte, the molar concentration of bis-carbonate could be obtained.

2.3.2. GC/MS analysis for gaseous degradation products study

EL-CELL®-type cells (ECC-Air model) were assembled in argon filled dry box, with 16.4 mg of the powder electrode described above (§2.2) as the negative electrode, and 2.5 cm² of the same commercial positive electrode. Two layers of Whatman GF/D borosilicate glass fibre were used as separator after impregnation of 300 µL of electrolyte. The powder electrode was lithiated/delithiated in different electrolytes. Each cell was connected to a pressure detector in order to follow gas generation during electrochemical treatment. After electrochemical treatment, the gas generated was sampled with an air-tight syringe. The sampled gas was directly injected to the GC/MS apparatus.

3. Results & discussions

The full cells assembled according to the experimental protocol were charged/discharged in different electrolytes based on EC and linear carbonate solvent mixtures. Then, the cells were opened for liquid GC/MS analysis, to follow principally the bis-carbonate generation stemming from the linear carbonate electrochemical reduction.

3.1. Ethylene bis-carbonate evolution in electrolytes based on EC and symmetrical linear carbonate solvent mixtures

3.1.1. Ethylene bis-carbonate product formation in EC/DEC (3/7 v/v)-LiPF₆ salt (1 M)

For the EC/DEC mixture, the corresponding bis-carbonate product 1' (Fig. 1 with R = C₂H₅) that is formed through steps i), ii) and iii), is mostly generated when the negative electrode reaches its lithiation plateaus (Fig. 2). There is a slight increase during the main electrolyte reduction process between 0.9 and 0.6 V but it remains insignificant compared to the maximum relative quantity of 1', being around thirty times smaller. Its relative quantity increases abruptly when the potential of the negative electrode reaches ~0.12 V.

Afterwards, a marginal increase is still observed during the lithium insertion plateaus, indicating that DEC reduction into ethoxide (Fig. 1) continues till it stabilizes around a maximum value of about 3.7 mol% vs. EC (0.17 mol.L⁻¹ with EC concentration at 4.5 mol.L⁻¹). This amount remains stable even after 50 cycles at C/50.

Note that the very first SEI layer formed during the solvent reduction process (between 0.9 and 0.6 V) does not have a good passivation property, as it still allows DEC reduction at low potential.

3.1.2. Ethylene bis-carbonate product formation in EC/DMC (1/1 w/w)-LiPF₆ salt (1 M)

Unlike the previous EC/DEC mixture, in EC/DMC, the corresponding degradation product 1₁ (Fig. 1 with R = CH₃) is mainly generated during the first reduction process that occurs between 0.9 and 0.6 V vs. Li⁺/Li⁰. Its amount is proportional to the quantity of electrons consumed during the process. This reduction process was conventionally assigned to EC reduction in the literature [21–23], since two decades ago, the main analytical technique to characterize it has been the direct observation of SEI layer through FTIR. 1₁ generation during the process indicates that it also involves DMC reduction giving methoxide and methyl formate radical (Fig. 1). This

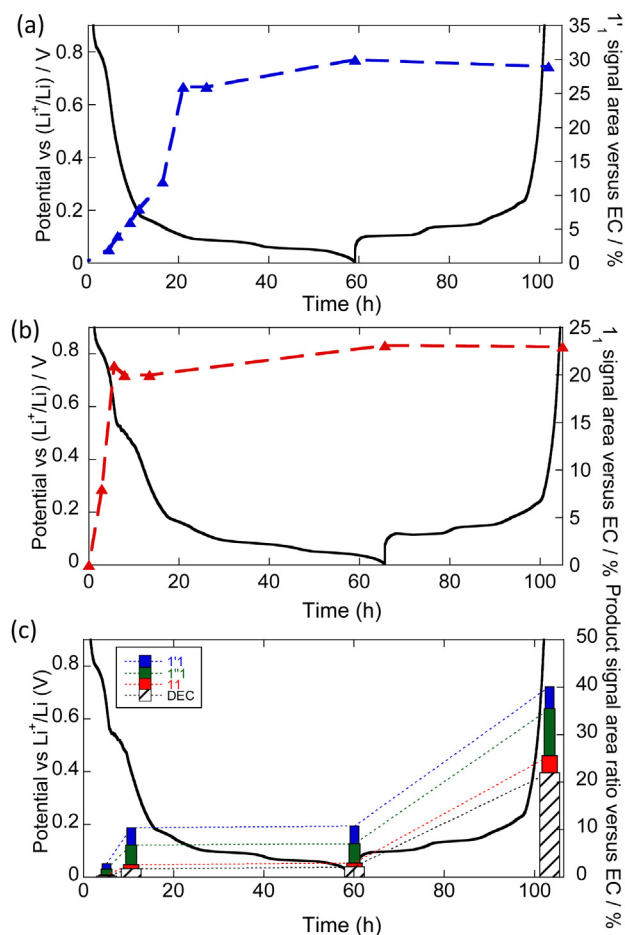


Fig. 2. Liquid GC/MS analysis of the electrolyte taken during the first cycle at C/50 in: (a) EC/DEC (1/1 w/w) - LiPF₆ salt (1 M); (b) EC/DMC (1/1 w/w) - LiPF₆ salt (1 M); (c) EC/DMC (1/1 w/w) - LiPF₆ salt (1 M). The electrochemical cycling curve is shown in black solid line. The evolutions in detected signal area vs. EC are given in blue for 1'1, red for 1 and green for 1'1.

is confirmed through methyl formate detection (from recombination of methyl formate radical with a hydrogen radical [10]) in the gas phase using GC/MS, just after the main reduction (0.6 V). The capacity consumed during this reduction of potential range between 0.9 and 0.6 V is about 40 (±8) mAh.g⁻¹ of Active Material (AM) which is higher than 26 (±4) mAh.g⁻¹ of AM consumed in the mixture EC/DEC. The supplementary electrons consumption may be due to DMC reduction.

Then, a marginal increase in the amount of bis-carbonate product is observed during the lithium insertion plateaus. The relative quantity of bis-carbonate vs. EC reaches a threshold value of about 2.8 mol% (0.2 mol.L⁻¹ with EC concentration at 6.4 mol.L⁻¹) and remains steady even after 50 cycles at C/50.

Note that a second reduction phenomenon is observed between 0.6 and 0.3 V. This phenomenon was already reported by Tao et al. as being a side reaction related to the intrinsic properties of graphite particles. It does not seem to have a noticeable effect over the alkyl bis-carbonate generation profile [24].

3.1.3. Discussion over the ethylene bis-carbonate generation profile in EC/DEC (3/7 v/v)-LiPF₆ salt (1 M) and in EC/DMC (1/1 w/w)-LiPF₆ salt (1 M)

The 1₁ and 1'1 generation profiles during the very first reduction/oxidation of graphite material in EC/DMC (1/1 w/w)-LiPF₆ salt (1 M) and EC/DEC (3/7 v/v)-LiPF₆ salt (1 M), respectively, highlights the different behaviours in both electrolytes. In EC/DMC, the

reduction potential range of 0.9–0.6 V is not surprising considering the reduction potential for each solvent in a single solvent system, where EC gets reduced at around 1 V (starting from ~ 1.05 V) and DMC mainly around 0.6 V (starting from ~ 1 V) (see supplementary information, fig. A). This suggests that the reduction phenomena observed in this mixture corresponds to a mixed potential between both solvents. In EC/DEC, the potential range for the main solvent reduction remains the same (0.9–0.6 V). The lower capacity consumed during this process can be explained by the DEC reduction potential that is around 0.5 V (starting from ~ 0.8 V) in a single solvent configuration, which is lower than for DMC. This may be explained by the stronger electron donating properties of the C_2H_5 -group compared to CH_3 -. In this case, DEC reduction giving ethoxide seems to occur preferentially during the lithium insertion plateaus.

The possible explanation for the difference observed between the two electrolytes comes from details observed through FTIR spectra in single solvent system during preliminary studies (see supplementary information, figs. C and D). The SEI obtained from DMC reduction is mainly composed of lithium methyl carbonate as well as lithium methoxide, in agreement with the literature [25]. Both of the products are generated from one-electron-step reduction of DMC, supporting that when DMC gets reduced, it consumes only one electron per molecule. While for the SEI obtained from DEC reduction, it mainly contains lithium carbonate (Li_2CO_3), supporting that DEC undergoes preferentially a two-electron-step reduction. Therefore, at relatively high potential (>0.2 V), DEC would more easily get reduced through a two-electron step, which could explain the low quantity of $1'_1$. At lower potential, it would rather get reduced through a one-electron step due to steric hindrance. Note that the one-electron step reduction product of DEC might not be limited to lithium ethoxide alone. Other molecules as EC should continue to get reduced to give lithium salt forming the SEI, since SEI growth was observed clearly at low potential [5].

In order to estimate the part of DMC getting reduced for methoxide formation, a semi-quantification using a chelating agent was conducted.

Table 1

Chemical simulation for 1_1 generation in different samples containing different amount of lithium methoxide in EC/DMC (1/1 w/w) - 1 M LiPF_6 .

Sample number	Total amount of MeOLi (10^{-2} mol/L)	Saturation	1_1 generated (10^{-2} mol/L)
1	15	Yes	22
2	5.54	Yes	21
3	2.32	Yes	22
4	1.79	Yes	22
5	0.79	No	22
6	0.45	No	22

3.1.4. Semi-quantification for electrochemical generation of alkoxide compounds

3.1.4.1. Attempt for direct quantification through chemical simulation. To quantify the amount of electrons consumed to form lithium methoxide in EC/DMC mixture through ethylene bis-carbonate compound quantification, a chemical simulation was conducted by adding different amount of lithium methoxide in the electrolyte. Lithium methoxide powder was added to EC/DMC (1/1 w/w)- LiPF_6 salt (1 M) electrolyte in argon-filled dry box.

As shown in Table 1, different amounts of lithium methoxide results in a same amount of bis-carbonate 1_1 that is around 0.2 mol/L^{-1} , whether or not there is saturation in lithium methoxide salt. It is worth noting that this concentration corresponds to the maximum value for 1_1 that was obtained after electrolyte degradation in battery condition (§3.1.2).

This result corroborates previous studies on bis-carbonate formation mechanisms that suggested the lithium methoxide regeneration [10]. According to the mechanism depicted in Fig. 1, alkoxide is not only responsible for bis-carbonate product formation, but it is also a by-product. Less than one equivalent of alkoxide is needed to form one equivalent of bis-carbonate, which makes impossible the direct quantification.

The fact that bis-carbonate concentration in the electrolyte remains fixed at any amount of lithium alkoxide in the system

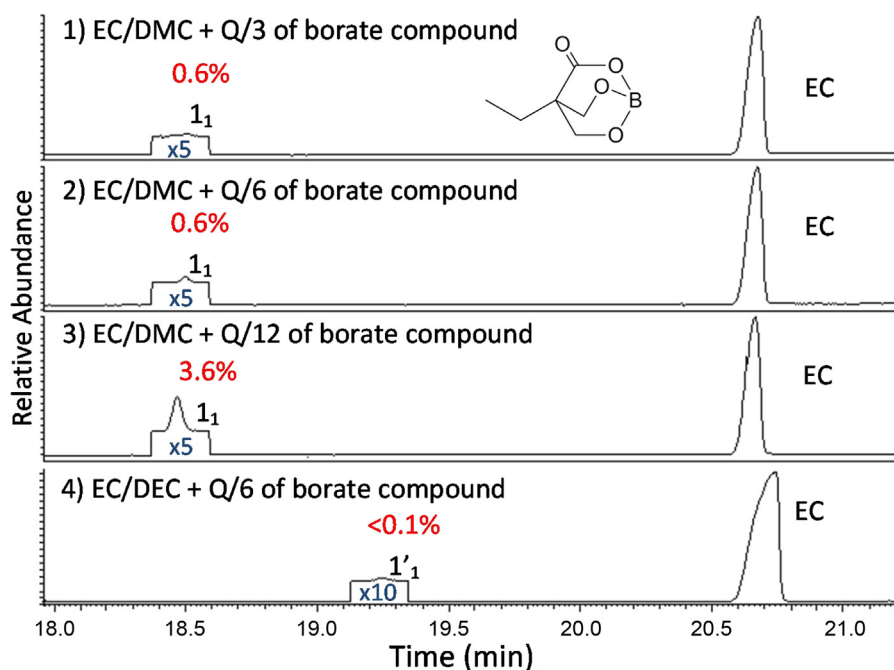


Fig. 3. Liquid GC/MS chromatograms of the electrolyte taken just after the main solvent reduction over graphite powder electrode in full cell configuration for (1), (2) and (3), and after one complete lithiation for (4). All electrolytes contained 1 M of LiPF_6 . Values in percentages give the integrated intensity of each peak compared to the integrated intensity of EC. The molecular formula of borate compound is given in inset.

proves also the reversibility for step ii) and iii). This reversibility was already reported in EC/DEC (1/1 v/v)-LiPF₆ salt (1 M). Sasaki and al. demonstrated that when lithium ethoxide is added to the electrolyte, it triggers 1'1 generation, and when it is added to 1'1 solution, there is EC and DEC formation [8].

3.1.4.2. Alkoxide quantification through chelating agent. Due to the alkoxide regeneration during bis-carbonate formation, it is impossible to quantify the amount of lithium alkoxide exclusively through bis-carbonate formation. In order to estimate the amount of lithium methoxide formed during the main reduction process, a boron-based chelating agent was used (Fig. 3). This compound is known to be a strong Lewis acid forming complexes with small anions such as O²⁻ and O₂²⁻ [20]. Complexation requires less energy than nucleophilic attacks where a new covalent bond is created. Therefore, if the complexing agent is strong enough, the nucleophilic attack entailing bis-carbonate product formation might be prevented.

In order to obtain the amount of methoxide generated during the main reduction process in EC/DMC (1/1 w/w) - LiPF₆ salt (1 M), the electrolytes with different borate concentrations (Q/3, Q/6, and Q/12, with Q the capacity consumed during the reduction process between 0.9 and 0.6 V in the same electrolyte that is equal to 40 mAhg⁻¹ of AM were analyzed through liquid GC/MS after the main solvent reduction process. The borate content does not influence the electrochemical behaviour of the electrolyte, and the capacity consumed during the main reduction remains unchanged, being around 38(±6) mAh.g⁻¹ of AM.

Fig. 3 shows that the minimum quantity of borate salt necessary to capture all the MeO⁻ is between 0.8 10⁻² mol.L⁻¹ (Q/6) and 0.4 10⁻² mol.L⁻¹ (Q/12). The first value -Q/6- can be considered as the maximum concentration required for preventing any bis-carbonate formation. Note that experimental value for lithium methoxide solubility in EC/DMC (1/1 w/w) - LiPF₆ salt (1 M)- is between 2.10⁻² mol.L⁻¹ and 0.8 10⁻² mol.L⁻¹. This value is in same order of magnitude with the solubility studies by Jones et al. [26] in pure solvents, where the methoxide solubility was evaluated for being around 1.7 10⁻² mol.L⁻¹ in EC, and around 0.24 10⁻² mol.L⁻¹ in DMC. The amount of methoxide scavenged by borate compound is equal or smaller than the amount soluble in the electrolyte. This supposes that the amount generated electrochemically during reduction between 0.9-0.6 V is small enough to be totally dissolved in the electrolyte, if the whole medium is homogeneous.

Same concentration of borate compound - Q/6 - was applied to EC/DEC (3/7 v/v)-LiPF₆ salt (1 M) - electrolyte. This amount, which corresponds to 1.5 wt% of the electrolyte, allows to stop efficiently the bis-carbonate product generation even after one complete cycle. This result proves that during the first cycle, the amount of lithium alkoxide formed in a mixture of EC/DEC is not more than the amount generated in EC/DMC after one complete lithiation.

3.2. Alkyl bis-carbonate evolution in electrolyte based on EC and asymmetric linear carbonate mixtures: bis-carbonate product formation in EC/EMC (1/1 w/w)- LiPF₆ salt (1 M)

After the influence of the two symmetric linear carbonates - DMC and DEC - on the bis-carbonate species generation profile during the first lithiation/delithiation of graphite anode was thoroughly checked, the influence of the most classical asymmetric linear carbonate - EMC - was investigated.

3.2.1. Evolution during the main solvent reduction process (0.9-0.6 V)

In EC/EMC (1/1 w/w) - LiPF₆ salt (1 M), several degradation products are observed right from the first reduction phenomena also

occurring between 0.9 and 0.6 V (Fig. 2c)). These products are DEC, DMC, 1₁, 1'1 and 1''1 (Fig. 4). The detected signals were low, being between 0.1 and 1.2% vs. EC. DMC vs. EC signal ratio could not be collected since DMC peak is covered by the acetonitrile solvent signal, but its presence could be detected through mass range spectra, and also through gas analysis. The detection of all the above mentioned degradation products is in agreement with the literature, where DEC and DMC were already observed through trans-esterification of EMC [13,27,28]. The three bis-carbonate compounds were also detected through liquid chromatography, after one complete cycle of a 400 mAh LiCoO₂/graphite full cell [9] with EC/EMC-based electrolyte. In their study, the first hypothesis concerning the bis-carbonate formation mechanism was emitted.

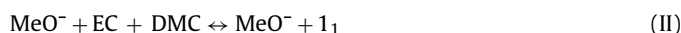
In this electrolyte, chemical degradation initiated by alkoxide nucleophilic attack can generate a huge diversity of compounds (Fig. 4), making the system far more complex than in the case of a mixture of EC with symmetric linear carbonate. These products all result from trans-esterification process between linear carbonates or bis-carbonates in the presence of alkoxide. Either ethoxide or methoxide is able to form all the degradation products. Yet, both methyl formate and ethyl formate are detected through gas analysis after one complete cycle (§3.1.2). Hence, EMC reduction seems to give both lithium methoxide and lithium ethoxide, and it is difficult to settle if one of them is generated first.

3.2.2. Evolution during the first graphite lithiation (0.6-0.005 V)

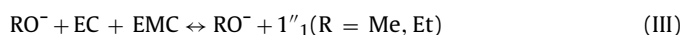
After a small increase in the relative quantities of the five degradation compounds reaching between 1 and 4% vs. EC in signal ratio, they stagnate afterwards at lower potential (Fig. 2c). After one complete lithiation, the amount of bis-carbonate in the electrolyte remains at least five times smaller than after the first complete lithiation in EC and symmetric linear carbonate mixture. For example, the relative intensity of the peak corresponding to 1'1 is at around 3.7% even after one complete charge whereas in EC/DEC mixture it reached 29%.

First of all, these low values can be explained by the huge amount of equilibriums where alkoxide is involved (Fig. 4). For example, for symmetric bis-carbonates such as 1₁ or 1'1, two global pathways exist for their generation. Especially for 1₁ formation, the pathways are:

(for 1'1, DEC instead of DMC, and MeO⁻ and EtO⁻ are inverted)



Or



In the first pathway ((I)+(II)), a certain amount of DMC is required to initiate the reaction (II). EMC trans-esterification is necessary for 1₁ formation, and this is why it is hindered compared to EC/DMC mixture, where DMC is already present in huge quantity.

In the second pathway ((III)+(IV)), a large amount of 1''1 has to be first formed to favor 1₁ formation through trans-esterification. But 1''1 formation requires two steps that are both reversible (Fig. 4 [vii-a) then vii-b) or vii-d]) or [viii-a) then viii-b) or viii-c])). Since EMC trans-esterification requires only one reversible step, it should be predominant to 1''1 formation.

On the other hand, no evolution is observed at lower potential, the lithium alkoxide might get trapped into the SEI layer, entrained by the lithium alkyl carbonate molecules of which the generation becomes predominant during lithium intercalation into graphite intercalation compounds (GIC) according to the literature [29].

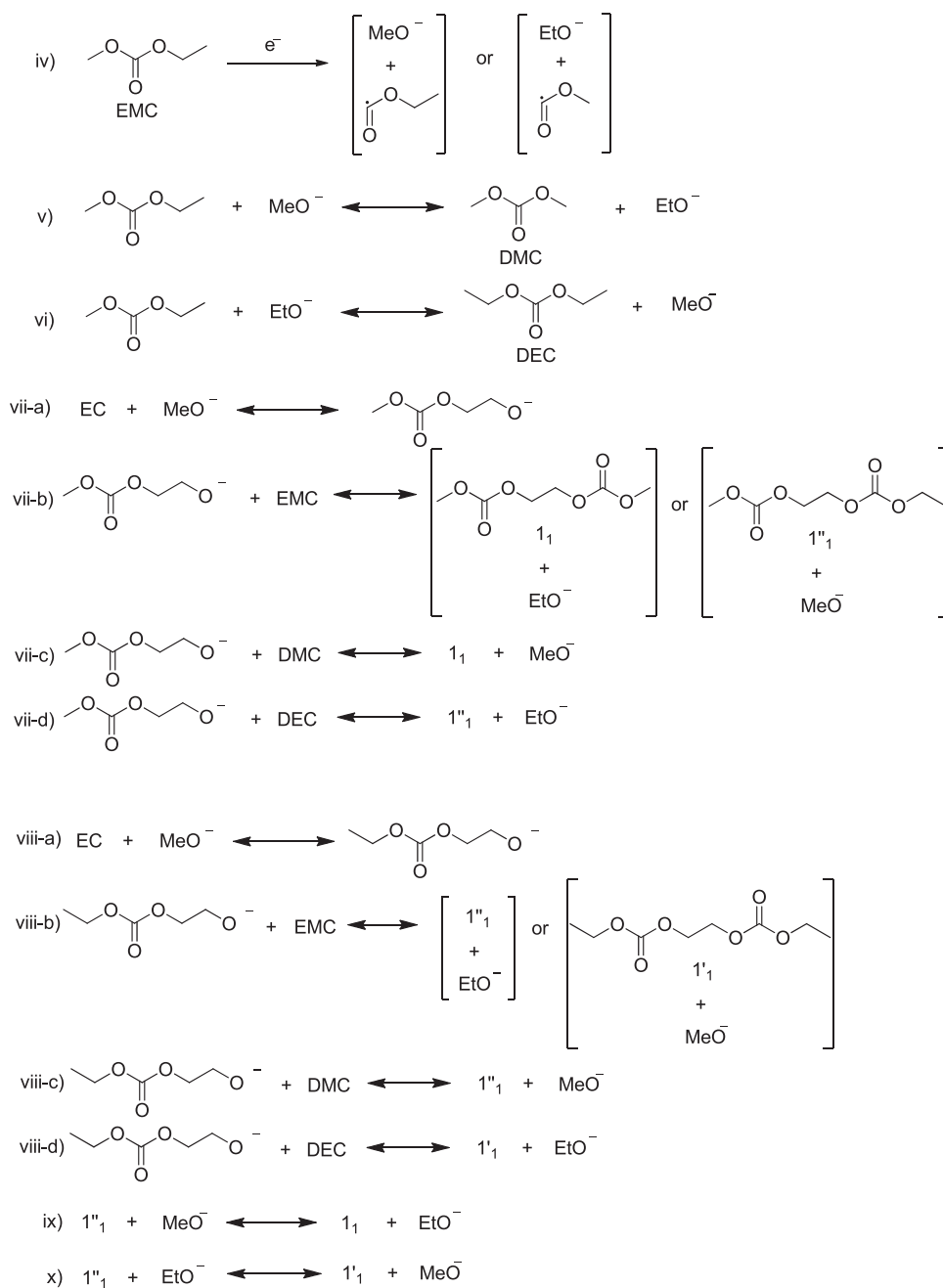


Fig. 4. Elementary steps for degradation under alkoxide action in mixture EC/EMC.

Moreover, the SEI layer compounds precipitate more easily as the interface is highly concentrated in lithium ions due to the overpotential for lithium insertion into graphite (Fig. 5a)). In case of EC/DEC, we observe the $1'_1$ generation up to 0.1 V (Fig. 2a), thus the precipitation of alkoxide seems less effective. This can be supported by studies in single solvents:

- Aurbach et al. [30] observed complete dissolution of metallic lithium in DEC.
- The first reduction curve in DEC-1 M LiPF₆ and EMC-1 M LiPF₆ shows that DEC is forming a SEI layer with less passivation ability (the lithium intercalation into graphite does not occur) than in the case of EMC (see fig. B, supplementary data). This is supported by the donor number of DEC which is higher than that of EMC [31], signifying that DEC is more favorable than EMC for salt dissolution.

3.2.3. Evolution during the graphite delithiation step

A large increase in bis-carbonate quantity occurs during the delithiation step, during which the carbonate solvent reduction cannot happen. The sudden availability of alkoxide compounds can only be explained by their release from the SEI, triggered by the current inversion within the system.

When the system shifts to delithiation step, the interface is no longer a zone with high concentration in lithium ions and anions (Fig. 5b). This increases the solubility of lithium salts at the interface, especially alkyl carbonates that have one of the highest solubility among the different SEI compounds [32]. Lithium alkyl carbonates at the outermost part of SEI gets dissolved at the interface, thus the electrolyte can access to lithium alkoxide compound and dissolve it.

On the other hand, in EC and symmetric linear carbonate mixtures, it is impossible to see if there is dissolution of SEI during

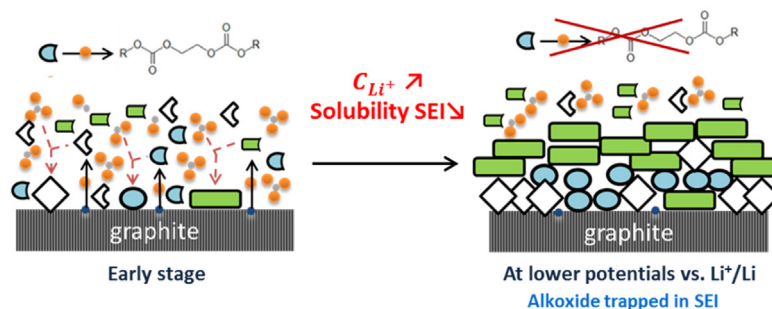
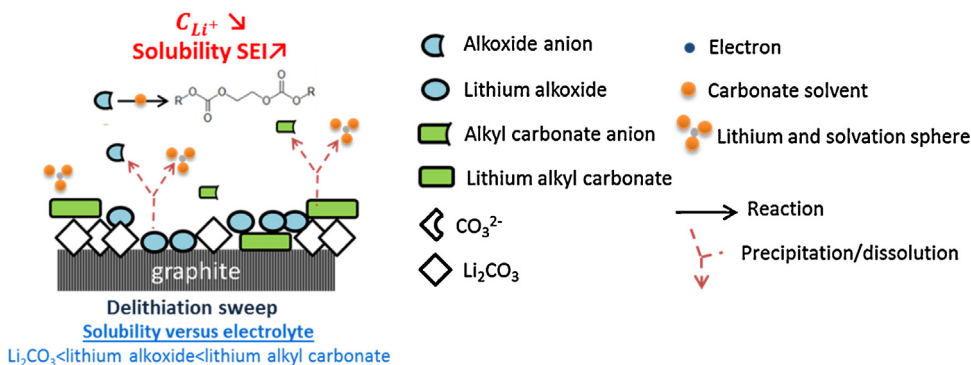
(a) SEI formation during the first lithiation step**(b) SEI dissolution during the first delithiation step**

Fig. 5. Schematic representation of SEI evolution during the first cycle in EC/EMC (1/1 w/w)- LiPF_6 salt (1 M) explaining how the soluble compounds are formed under alkoxide action.

the delithiation step through bis-carbonate compound generation since its concentration reaches the maximum threshold value from the very beginning of the lithiation step. However, evidences support that the dissolution of SEI is effective during the delithiation step in a carbonate-based electrolyte:

- A preliminary study using FTIR on the negative electrode surface after a delithiation to 3 V in EC/DMC (1/1 w/w)- LiPF_6 salt (1 M) in a

half cell configuration shows that the surface is mainly composed of lithium carbonate, whereas after a discharge to 1.5 V, lithium alkyl carbonate is also detected.

- In literature, a study using electrochemical quartz crystal microbalance associated to platinum electrode corroborates the SEI dissolution during discharge step in a carbonate-based electrolyte using LiPF_6 salt, since a negative mass change with current passed in oxidation was observed [29]. Also, a study using TEM

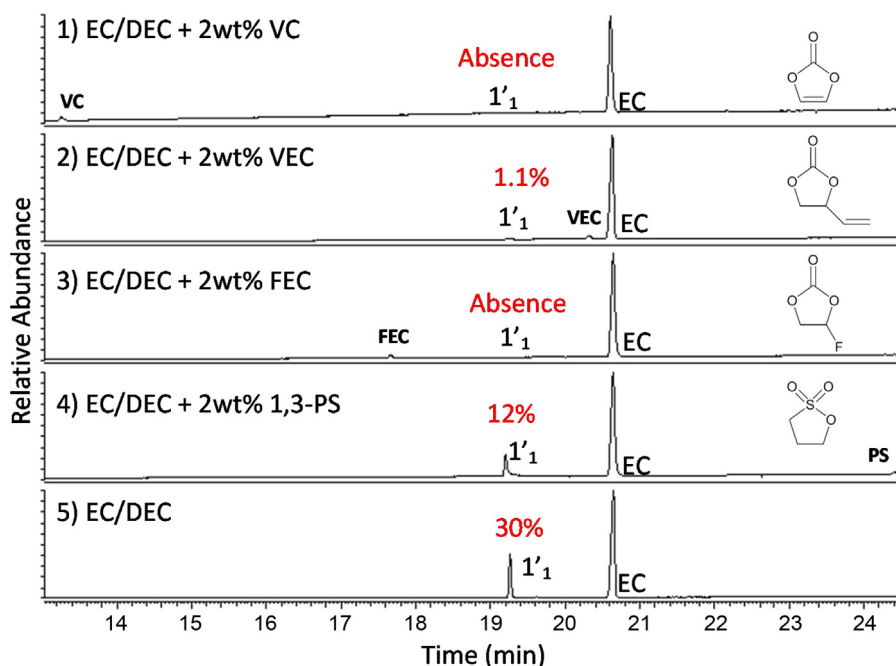


Fig. 6. Liquid GC/MS analysis after one complete lithiation of graphite powder electrode in full cell configuration. All the electrolytes contain 1 M of LiPF_6 . Values in percentages give the integrated intensity of each peak compared to the integrated intensity of EC.

over a conversion type anode showed a thinner SEI layer after one cycle than in the completely lithiated state [33].

3.3. Use of additives for more passivating SEI

The bis-carbonate compounds are known to decrease the conductivity of the electrolyte and generate a very rigid SEI through reduction [12] and thus hinder good cyclability for batteries. Therefore, preventing their formation can only have beneficial effect over the cycle life of a battery.

Four of the most commonly used additives in the literature –VC, VEC, FEC and 1,3-PS– were chosen and added to EC/DEC (3/7 v/v)–LiPF₆ salt (1 M) electrolyte at a concentration of 2 wt%, to see if they can also prevent bis-carbonate formation even at an early stage of degradation. These additives are known to get reduced before EC or linear carbonates according to the LUMO prediction [34], and form a more stable SEI.

Fig. 6 shows that all carbonate additives –VC, VEC and FEC – at 2 wt% inhibit the bis-carbonate product formation. Even after one complete lithiation of graphite anode, the bis-carbonate products were detected at less than 1% vs. EC signal. No ethyl formate was detected in the gas collected analysed through GC/MS, which confirms that DEC reduction into ethoxide is nearly nonexistent.

The effect of VC over bis-carbonate product inhibition has been already reported in the literature. The main explanation was that the RO[–] anion undergoes preferential nucleophilic attack on VC to form alkoxy ethylene carbonate (AEC), and this product can be detected through GC/MS [11]. Since neither EC nor DEC reduction is observed with addition of VC, and that AEC is not detected during post-mortem analysis, it is more likely that the SEI formed from VC reduction has excellent passivation properties and thus prevents DEC from getting reduced to form EtO[–]. A similar process is speculated for FEC and VEC, although the passivation layer formed from VEC degradation product seems less effective since a small amount of bis-carbonate product is still detectable.

The first SEI layer obtained by 1,3-PS reduction displays less good passivation properties than the three other carbonate-based additives, because bis-carbonate could be still detected with a relative signal intensity of 12% vs. EC after one complete lithiation. Yet, the signal remains smaller than with the electrolyte without any additives.

4. Conclusion

The present study highlights the role of linear carbonates in the early stage of electrolyte degradation process for lithium-ion batteries. Specifically, it shows that DMC and EMC reduction giving alkoxide takes place during the main electrolyte reduction peak, whereas for DEC, it occurs at lower potential. This early reduction process is responsible for the generation of bis-carbonates that are soluble in electrolyte and can be detected through GC/MS. These soluble degradation compounds can provide information on the trans-esterification process, the chemical degradation reactions and the SEI formation process, including the dissolution of SEI compounds during discharge. The soluble degradation compounds can act as a probe for evaluating the SEI layer passivation properties while using SEI reinforcing additives such as VC, VEC, FEC and 1,3-PS. The freshly formed SEI layer from carbonate-based additives was shown to have better passivation than the one obtained from 1,3-PS. Hence, this approach can be applied to discriminate battery formation parameters such as the charging protocols, the relaxation period, and temperature, or to examine the influence of the electrode surface area and water content to SEI layer formation, and also the impact of manganese dissolution to supplementary electrolyte reduction.

This paper also emphasizes that the borate compound, when introduced to complex lithium alkoxides, can be an interesting additive for electrolyte. It blocks the chemical degradation of the electrolyte coming from alkoxide and it can have a beneficial effect on the overall cycle life of a battery.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.electacta.2014.05.072>.

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